

Safety culture and nuclear safety management. Analysis of the relationships of its variables through a bibliographic review and AHP.

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Objective: Based on previous literature reviews, this study examines the evidence about the key elements of the safety culture constructs applied in nuclear reactor operating organizations. **Background:** Safety culture (SC) is an organizational concept born in the nuclear industry to the behavioural elements in the safety management of nuclear facilities. This concept has been continuously revisited by academics and practitioners and has even been widely adopted by the conventional industry. **Methods:** systematic bibliographic review to identify key concepts of safety culture. Analytical hierarchy process (AHP) is applied to rank these concepts through an online questionnaire delivered to experts; **Results:** six articles were found, and four key concepts of safety culture were identified. These concepts were ranked in the following order after expert AHP online questionnaire results analysis: top management leadership, communication management, safety climate, and hazard and risk analysis; **Conclusions:** Based on this research the leadership actions are the most important nuclear safety management action to achieve the nuclear safety goals of the nuclear reactor operation.

Keywords: nuclear, safety, management, reactors, AHP.

1. Introduction

Safety culture (SC) is an organizational concept born in the nuclear industry to the behavioural elements in the safety management of nuclear facilities. Proposed by the International Nuclear Safety Advisory Group (INSAG) in (INSAG 1986) and improved in (INSAG 1991), this concept has been continuously revisited by academics and practitioners and has even been widely adopted by the conventional industry.

This original state defines: “Safety Culture is that assembly of characteristics and attitudes in organizations and individuals which establishes that, as an overriding priority, nuclear plant safety issues receive the attention warranted by their significance” (INSAG 1991).

The Analytic Hierarchy Process (AHP) (Saaty 1990) is a methodology to model multiple criteria decision-making problems. The methodology process enables the quantification of subjective elements through the paired criteria comparison mediated by decision criteria. The nuclear industry has adopted the AHP methodology from the first years that this was elaborated. The first document was published in 1996 (Holmberg and Pulkkinen 1996). AHP has been implemented fundamentally for the Nuclear Safety area as a multi-criteria decision-making support tool. Its main contribution is carrying out safety analysis of nuclear power reactors, evaluating potential emergencies and their consequences, and analysing human reliability. Regarding AHP applied to safety culture evaluation in (Ali 2016), (Situmorang et al. 2018), (Felice et al. 2016),

AHP was used to determine objectives, identify organizational precursors to accidents, and prioritize implementation of safety management practices.

2. Methodology

A systematic review of the literature based on (José de Oliveira et al. 2019) using SCOPUS and INIS identifies the key variables of safety culture. Then, the AHP method of (Saaty 1990) is used to rank the relative importance of the identified variables. The input to the AHP methodology is obtained in two stages.

First, through an online questionnaire based on (Bryman and Bell 2011). In this questionnaire, experts' preferences between variables identified in the systematic review of the literature were investigated. The experts included researchers, nuclear reactor operators, and nuclear industry regulators.

Then, in the second stage, the preferences of the research team are surveyed in the comparison of all the criteria postulated and not repeated. In this way, the judgment of experts is obtained for the order of the variables and the judgment of the researchers to sensitize in future investigations to the relative importance between the postulated criteria.

Finally, the results are calculated, processed, and interpreted using the AHP methodology.

3. Research

3.1. Identification of safety culture key concepts (SCKC).

Safety culture (SC) is an organizational concept typical of nuclear safety applied in the management activities of nuclear reactors (Bernard 2018). Since its formulation in (INSAG 1991), this construct has received many different approaches and contributions, moving away from the original approach. In that sense, (Guldenmund 2010) stands out which saying, "the culture concept is deprived of much of its depth and subtlety and is morphed into a grab bag of behavioural and other visible characteristics, without reference to the meaning these characteristics might have and often infused with normative overtones."

Based on this observation, this work considers the various reviews of the state-of-the-art and others bibliographic reviews of safety culture to identify the most common and relevant concepts attributed to the primary and original construct. Using SCOPUS and INIS and Using the following search strings in "abstract, title and keywords": "safety culture" AND (review OR state-of-the-art OR bibliometric") after removing duplicates, six documents were recovered. After obtaining the documents, content analysis was carried out to identify key concepts in the SC. The content analysis implied recognizing the most relevant concepts regarding SC in each work. The papers that were retrieved and the SC key concepts identified are listed in Table 1.

Table 1 Relevant SC bibliographic reviews papers and their SC key concepts.

#	Paper	Period conside red	SC Key Concepts
1	(Guldenmund 2000)	1980-2000	Safety climate Management commitment, safety training, open communications, environmental control and management, stable workforce, positive safety promotion policy
2	(Sorensen 2002)	1980-2000	Safety climate, safety management systems Behavioural practices, norms, for anthropological: beliefs, values, attitudes, assumptions, expectations
3	(Choudhry, Fang, and Mohamed 2007)	1998-2007	Safety performance, top leadership management, risk management.
4	(Edwards, Davey, and Armstrong 2013)	1983-2010	
5	(Dominic Cooper 2016)	1991-2016	

6	(Bernard 2018)	2008-2018	Safety management system, learning, leadership, human performance
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After analyzing the works presented in Table 1, it is observed that the key concepts most addressed, mentioned and surveyed in the content of papers analyzed are: safety climate (Guldenmund 2000), (Choudhry, Fang, and Mohamed 2007), top leadership management (Bernard 2018), (Domnic Cooper 2016) y safety management systems (SMS) (Bernard 2018), (Choudhry, Fang, and Mohamed 2007).

3.2. Concepts operationalization.

This section presents the operationalization of the SC key concepts identified in previous literature/research (safety climate, top leadership management, and SMS). Each concept is shown in the context of practical and pragmatic SC and safety management. Each concept is presented with an operational definition. In the case of SMS, it took a particular operationalization considering two of the more distinctive process of this type of system: hazard and risk analysis and communication management (Acuña, Brollo, and Torres 2019), (Domnic Cooper 2016).

In this sense, communication management is a managerial practice of effective transmission of information within the organization, which influences and directs people's behavior (Newstrom 2011). On the other, hazard and risk analysis is a decision-making process characteristic of the nuclear industry, which is guided by regulations that are mostly risk-informed. For this reason, this typical process in the SMS of said industry gives specific input to the decisions and actions of senior management and, consequently, to its exercise of leadership on the safety status of the nuclear reactor organization. Next, the concepts are defined as variables by the research team of this paper.

Safety climate (V1): global perception of the safety aspects of the organization that can serve as a reference on which to develop one's own safe/unsafe behavior or on the judgment of others' patterns of safe/unsafe behavior. Top

management leadership (V2): set of features and actions of the organization's senior management to achieve individual and organizational radiation and nuclear safety objectives. Hazard and risk analysis (V3): the process of identification and evaluation of the dangers of activity, the conditions that originate them, and their potential unwanted consequences. Communication management (V4): set of actions and processes (of design, planning, execution, measurement, and correction) to make known and knowing news pertinent to the nuclear safety of the organization.

3.2 AHP problem structure.

The AHP problem is structured by the objective of rank the impact on safety management of the SC elements identified in the previous section. For this purpose following (Saaty 1990), three comparison criteria are defined for the four variables (SCKC identified in section 3.1). Thus, the structure of the problem is designed to compare for each criterion the relative importance between all the possible and non-repeated pairs of variables, the numerical analysis of the results is carried out by applying matrix algebra.

The mentioned criteria are: Criteria 1 (C1), general impact: Impact of the element on the general safety management results (achievement of organization safety goals). In this case, this criterion is used intended to determine which elements are the most important considering their overall impact on the organization safety goals and, considering the different elements' synergies. Criteria 2 (C2), individual contribution: impact of the individual (quantitative/qualitative) value of the element on the achievement of SMS safety objectives. Criteria 3 (C3), impact between elements: how much impact has one element on the other. This criterion is intended to determine which elements the most important are considering their mutual impact. In other words what element impact in the achievement of organization safety goals more than others?

The C1 criteria responds to evaluating the relevance of each concept in achieving the nuclear safety objectives of the facility. The C2 criteria allows evaluating the individual impact of the concept without considering that its manifestation

in management interacts with the result of other concepts. The C3 allows evaluating the individual impact considering synergies or feedbacks of the performance of the other elements. In this way, C2 and C3 cover the possible dynamics that could occur in operational management and enable the expert who answers the questionnaire to evaluate relative importance.

With C1, C2, C3, and the concepts operationalized into variables the AHP problem and its structure is defined (see **¡Error! No se encuentra el origen de la referencia.**). Based on the structure of the problem, an online questionnaire was designed to collect the necessary data to perform the calculations required by the AHP methodology.

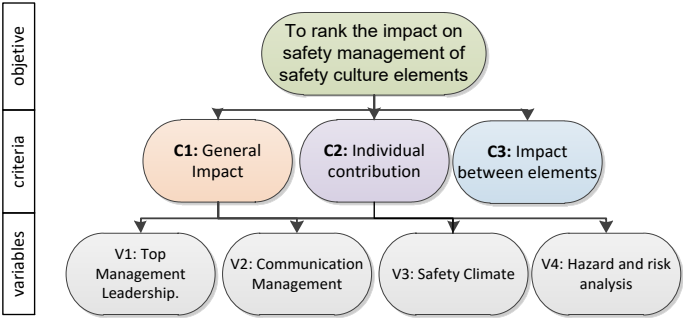


Fig. 1 AHP problem structure.

3.4 Questionnaire design

Based on the structure of the problem, a questionnaire was designed to collect the necessary data to perform the calculations required by the AHP methodology (Saaty 1990). The questionnaire consisted in 18 questions structured in three sections (one for each criteria) of six questions each one. In each section, the expert was asked to indicate his preference between two of variables postulated covering the possible combinations of V1, V2, V3, and V4.

The definition of each variable was included in each question. To reveal the preference between the variables was used a Likert scale (0-9: 1 equal importance, 3 somewhat more important, 5 much more important, 7 very much more important, 9 absolutely more important and 2,4,6,8 like intermediate values). The definitions necessary to answer it are provided for each question.

The questionnaire was elaborated and was accessible via Google Forms (see <https://forms.gle/5f4axLHAhGQWi9Qn7>).

3.5 Results.

The questionnaire was send via mail and Research Gate to the mentioned experts.

Twenty questionnaires with responses were received. The distribution of responses according the type of experts was: academics (identified in papers related to SC research): 60% of total responses, middle managers and nuclear reactor (PHWR and CANDU) operators: 20 % of total responses, regulators: 20 % of total responses.

The results were processed using matrix algebra following (Saaty 1990) to obtain their normalized matrices, eigenvectors, and eigenvalues for each section of the questionnaire according to the AHP methodology. The results were evaluated individually by the consistency ratio (CR) and the consistency index (CI). Only the responses with a $CR < 0.1$ were accepted. With these results were calculated an arithmetic average of the accepted matrix responses considering 14 questionnaire responses (20% of the researchers questionnaire responses, 5 % of middle managers and nuclear reactor operators questionnaire responses and 5% of regulators questionnaire responses were rejected for not meeting the acceptance criteria of the CR). To improve this approach future research could consider geometric averages.

For the aggregated analysis of the matrixes for the calculation of the arithmetic averages, the acceptance criteria of CR was increased to 0.28 according the dispersion of the values and the considerations presented in (Brufman 2015).

These results are shown in Table 2, Table 3 and Table 4.

Table 2 Calculation of preferences for C1.

Variables	V1	V2	V3	V4	Normalized matrix				Eigenvector
V1	1	5	6	6	0.64	0.79	0.44	0.36	0.56
V2	1/5	1	6	5	0.13	0.15	0.46	0.32	0.27
V3	1/6	1/6	1	4	0.12	0.03	0.08	0.25	0.12
V4	1/6	1/5	1/4	1	0.11	0.03	0.02	0.06	0.06
TOTAL	1.55	6.48	12.70	15.56	-	-	-	-	-

Table 3 Calculation of preferences for C2.

Variables	V1	V2	V3	V4	Normalized matrix				Eigenvector
V1	1	6	6	6	0.64	0.93	0.49	0.31	0.59
V2	1/6	1	4	5	0.11	0.15	0.35	0.26	0.22
V3	1/6	2/9	1	4	0.10	0.03	0.08	0.20	0.10
V4	1/5	1/4	1/3	1	0.13	0.04	0.03	0.06	0.06
TOTAL	1.55	6.48	12.70	15.56	-	-	-	-	-

Table 4 Calculation of preferences for C3.

Variables	V1	V2	V3	V4	Normalized matrix				Eigenvector
V1	1	6	7	6	0.64	0.96	0.52	0.36	0.62
V2	1/6	1	6	5	0.10	0.15	0.46	0.29	0.25
V3	1/7	1/6	1	4	0.10	0.03	0.08	0.26	0.12
V4	1/6	2/9	1/4	1	0.11	0.03	0.02	0.06	0.06
TOTAL	1.49	7.61	13.69	15.33	-	-	-	-	-

The research team elaborated the pairwise comparison between criteria's (C1, C2 and C3). The results are shown in Table 5 and shown that C2 (Individual contribution) it is the most relevant criteria followed by C3 (Impact between elements) and C2 (General impact).

Table 5 Weighting of the criteria

Criteria	C1	C2	C3	Normalized matrix				Eigenvector	Research team prioritization
C2	9	1	1	0.53	0.53	0.47	0.47	0.50	1°
C3	7	1	1	0.41	0.41	0.47	0.47	0.44	2°
C1	1	1/9	1/7	0.06	0.06	0.05	0.07	0.06	3°
TOTAL	17.00	2.11	2.14	-	-	-	-	-	-

According to the previous statements, the results obtained of the prioritization are shown in Table 6.

Table 6 Results

Variables/Criteria	C1	C2	C3	Total	Priorization
Top management leadership	1.55	1.53	1.49	1.51	1°
Communication management	6.48	7.47	7.61	7.47	2°
Safety climate	12.7	11.99	13.69	12.78	3°
Hazard and risk analysis	15.56	13.11	15.33	14.23	4°
Weighting of the criteria	0.06	0.50	0.44	-	-

4. Conclusions

The results obtained (priority of leadership practices (variable V1) over the other variables) seem to be in accordance with the most recent recommendations or safety requirements and recommendations of international organizations. Both the International Atomic Energy Agency (IAEA) (IAEA 2016), the World Association of Nuclear Operators (WANO) (WANO 2016) and the Nuclear Energy Atomic (NEA) (OECD/NEA 2017) have promoted pragmatic leadership actions through practical tools such as SMS and training for plant personnel. Possibly this may have induced a concrete internalization of abstract safety management concepts such as leadership in day-to-day operational practice in nuclear reactors or plants. In this way, the most sensitive to this variable (V1) are both the middle managers and reactor operators, as well as the researchers whose lines of investigation are in contact with them, are also.

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